

Integrated Credit Customer Strategy

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1.1 Background

The modern credit landscape is characterized by increasing competition, diverse customer profiles, and heightened regulatory scrutiny. Financial institutions face persistent pressure to balance growth objectives with prudent risk management. The expansion of digital banking and the growing availability of customer-level data have enabled lenders to access increasingly granular demographic, behavioral, and transactional information.

These developments have facilitated the adoption of predictive analytics in credit risk management. By leveraging advanced modeling techniques, institutions can move beyond traditional credit assessment methods that rely primarily on static demographic or income-based indicators, and instead adopt dynamic, behavior-driven approaches to credit evaluation.

Machine learning-based credit scoring models, in particular, enable more precise estimation of default risk. However, predictive accuracy alone is insufficient to generate sustained business value. For such models to be effective in practice, they must be embedded within decision frameworks that translate risk predictions into operational actions, specifically, the alignment of credit limits, product offerings, and pricing structures with borrower risk profiles.

1.2 Problem Statement

Despite significant advances in credit risk modeling, many financial institutions continue to face persistent challenges in operationalizing predictive insights. Three recurring issues are particularly notable:

1. **Misalignment between predicted risk and product allocation:** Low-risk customers may be under-served through conservative limits or unsuitable products, while high-risk borrowers may be extended excessive credit exposure, resulting in suboptimal portfolio performance.
2. **Inefficient pricing strategies:** Interest rates and product features are frequently determined without fully incorporating the heterogeneity of borrower risk, leading either to elevated credit losses or forgone revenue opportunities.
3. **Limited operationalization of predictive models:** Even highly accurate credit scoring models often fail to inform actionable eligibility criteria, product recommendations, or risk-tiered pricing strategies that can guide day-to-day lending decisions.

This project addresses these challenges by developing an integrated credit scoring and eligibility framework that translates predictive risk estimates into business-ready policies. The framework leverages borrower behavior, predicted default probabilities, and utilization metrics to inform credit limits, product assignment, and pricing decisions.

1.3 Objectives

The primary objectives of this project are as follows:

1. To develop a predictive credit scoring model capable of estimating the likelihood of default using customer demographic, behavioral, and transactional data.
2. To segment the customer portfolio based on risk, utilization, and repayment behavior, enabling differentiated treatment of borrowers.
3. To design an integrated eligibility framework that aligns predicted risk with appropriate product offerings and pricing tiers.
4. To optimize portfolio management strategies by constraining high-risk exposure while maximizing growth and profitability among low-risk customers.
5. To generate actionable business insights to support both operational decision-making and strategic planning.

1.4 Methodology

This study utilizes a publicly available credit card client dataset that contains detailed demographic characteristics, financial attributes, and repayment behavior. The data were obtained from the UCI Machine Learning Repository and are available at: [UCI Credit Card Dataset](#)

Methodologically, the framework is organized into four interlocking analytical layers:

1. **Descriptive and Diagnostic Analytics:** To explore default behavior, portfolio composition, and the structural drivers of credit risk across customer segments.
2. **Behavioral Feature Engineering:** To transform raw transactional and repayment data into economically meaningful indicators of credit risk, capturing patterns of utilization, payment discipline, and repayment delays.
3. **Unsupervised Customer Segmentation:** To identify heterogeneity in credit usage and repayment behavior by clustering borrowers into distinct segments with shared risk and behavioral characteristics.
4. **Supervised Credit Risk Modeling (H2O AutoML):** To estimate individual-level default probabilities using machine learning models and to translate these predictions into operational credit decisions within an integrated eligibility and pricing framework.

1.1 Data Source and Initial Processing

The data were imported from an Excel source using the *readxl* package in R.

```
# =====  
# LOAD DATA  
# =====  
df <- read_excel("default of credit card clients.xls", skip = 1)  
df <- as.data.frame(df)  
  
# Rename target variable for convenience  
df$default_payment_next_month <- df$`default payment next month`  
df$`default payment next month` <- NULL
```

The target variable, originally labeled default payment next month, was renamed to “default_payment_next_month”. Observations represent individual credit card holders, with demographic attributes, credit limits, billing amounts, repayment history, and default outcomes.

1.2 Data Cleaning and Feature Engineering

Categorical Encoding and Risk Flag Construction

Key demographic variables such as gender, education, and marital status, were transformed into interpretable factor variables. Age was discretized into economically meaningful cohorts to capture life-cycle effects in credit behavior. A binary risk flag was introduced to distinguish defaulting and non-defaulting customers at the exploratory stage.

```
df_clean <- df %>%  
  mutate(  
    SEX = factor(SEX, labels = c("Male", "Female")),  
    EDUCATION = factor(EDUCATION,  
                        levels = c(1, 2, 3, 4),  
                        labels = c("Graduate School", "University", "High School", "Others")),  
    MARRIAGE = factor(MARRIAGE,  
                      levels = c(1, 2, 3),  
                      labels = c("Married", "Single", "Others")),  
    AGE_BIN = cut(AGE, breaks = c(18, 30, 40, 50, 60, 100),  
                 labels = c("18-30", "31-40", "41-50", "51-60", "60+")),  
    RISK_FLAG = ifelse(default_payment_next_month == 1, "High Risk", "Low Risk")  
  )
```

Incomplete observations were removed to ensure statistical consistency:

```
# Removing NaNs  
df_clean <- df_clean[complete.cases(df_clean), ]
```

2. Exploratory Data Analysis (EDA)

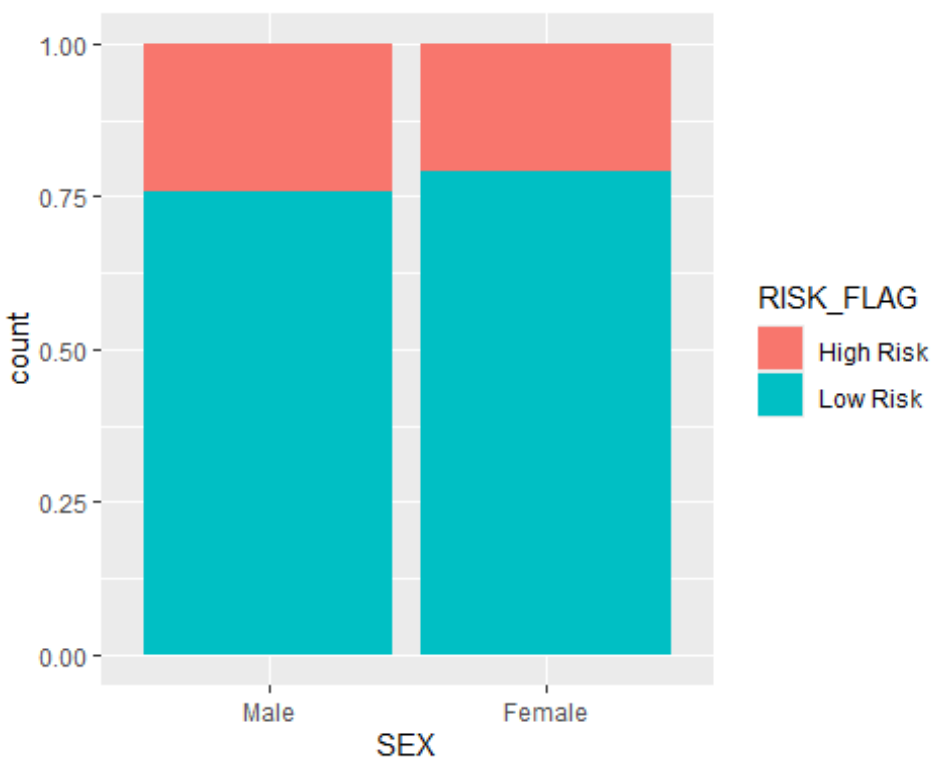
Exploratory Data Analysis (EDA) served two primary objectives:

- diagnosing the structural distribution of credit risk across the portfolio, and
- validating economic intuition prior to model construction.

2.1 Default Distribution and Demographic Risk Patterns

Default rates were examined across key demographic dimensions, including gender, age groups, and marital status, using proportionate bar charts. This visual analysis enabled an assessment of demographic risk gradients and helped ensure that observed patterns reflected economically plausible relationships.

```
eda <- df_clean
ggplot(eda, aes(x = SEX, fill = RISK_FLAG)) +
  geom_bar(position = "fill")
```



2.2 Payment History and Behavioral Risk

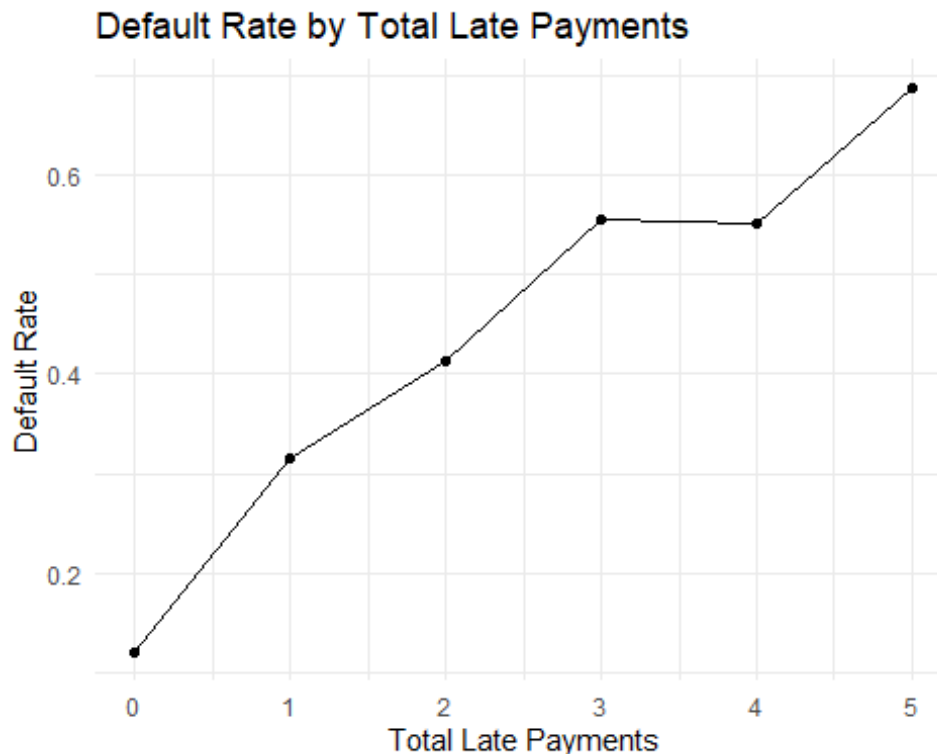
Repayment behavior is central to credit risk. Late payment indicators (PAY_0 to PAY_5) were aggregated into: - Total late payments - Average delay severity

```
pay_cols <- c("PAY_0", "PAY_2", "PAY_3", "PAY_4", "PAY_5")
eda <- eda %>%
  mutate(
```

```

    total_late_payments = rowSums(select(., all_of(pay_cols)) > 0),
    avg_late_payment = rowMeans(select(., all_of(pay_cols)))
  )
eda %>%
  group_by(total_late_payments) %>%
  summarise(default_rate = mean(default_payment_next_month)) %>%
  ggplot(aes(x = total_late_payments, y = default_rate)) +
  geom_line(group = 1) +
  geom_point() +
  labs(
    title = "Default Rate by Total Late Payments",
    x = "Total Late Payments",
    y = "Default Rate"
  ) +
  theme_minimal()

```



A monotonic relationship between cumulative late payments and default probability was empirically validated.

2.3 Correlation and Dependency Analysis

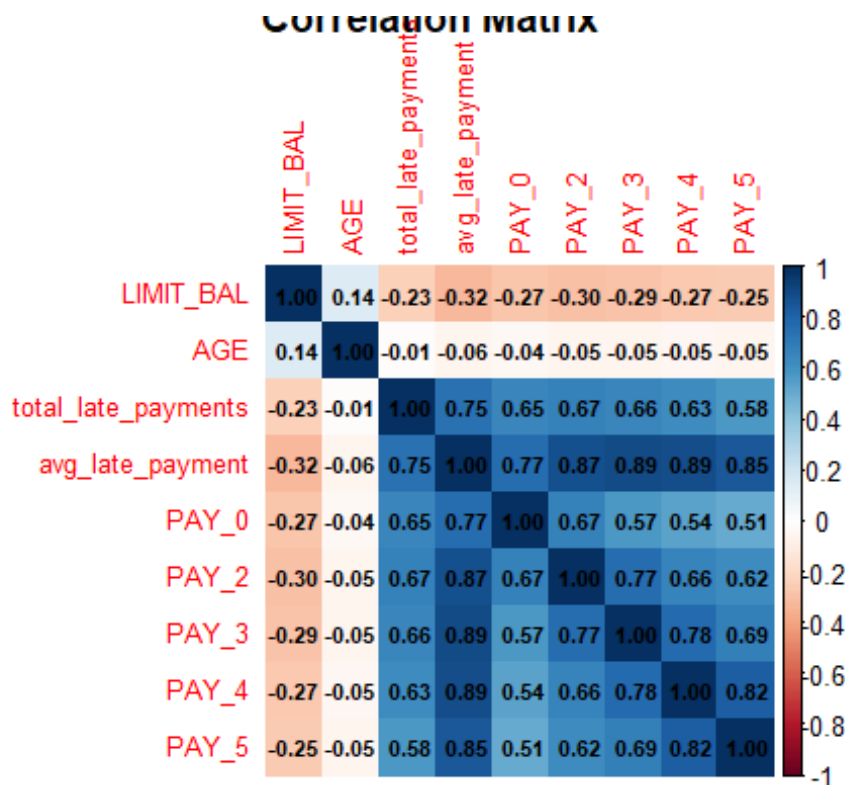
To assess linear dependencies and potential multicollinearity among numeric predictors, a correlation matrix was computed and visualized:

```

num_cols <- c("LIMIT_BAL", "AGE", "total_late_payments", "avg_late_payment", pay_cols)
corr_matrix <- cor(eda[, num_cols])

```

```
corrplot(corr_matrix, method="color", addCoef.col="black", number.cex=0.7, tl
.cex=0.8, title="Correlation Matrix")
```



This step informed later modeling choices by identifying highly correlated billing and payment variables, which were subsequently consolidated into summary indicators.

2.4 Business-Oriented Segment Analysis

Recognizing that risk is not homogeneous across customers, segmentation analyses were conducted along economically salient dimensions.

2.4.1 Credit Limit Stratification

Customers were grouped into quartiles based on credit limit size to examine scale effects in default behavior:

```
eda <- eda %>%
  mutate(LIMIT_QUARTILE = ntile(LIMIT_BAL, 4))
```

2.5 Advanced Behavioral Feature Construction

To improve predictive signal quality, raw transactional variables were transformed into economically interpretable aggregates: - Average payment delay - Average bill amount - Average payment amount - Credit utilization ratio

```
df_clean <- df_clean %>%
  mutate(
```

```

    AVG_PAY_DELAY = rowMeans(select(., PAY_0:PAY_6)),
    AVG_BILL_AMT   = rowMeans(select(., BILL_AMT1:BILL_AMT6)),
    AVG_PAY_AMT    = rowMeans(select(., PAY_AMT1:PAY_AMT6)),
    UTILIZATION    = AVG_BILL_AMT / LIMIT_BAL
  )

# Model dataset
df_model <- df_clean %>%
  select(LIMIT_BAL, AGE, AVG_PAY_DELAY, AVG_PAY_AMT, AVG_BILL_AMT, UTILIZATION) %>%
  mutate_all(~replace(., is.na(.), 0))

df_scaled <- scale(df_model)

```

3. Customer Segmentation via K-Means Clustering

Unsupervised segmentation was performed using scaled behavioral features. The optimal number of clusters was selected via the elbow method, balancing parsimony and interpretability.

```

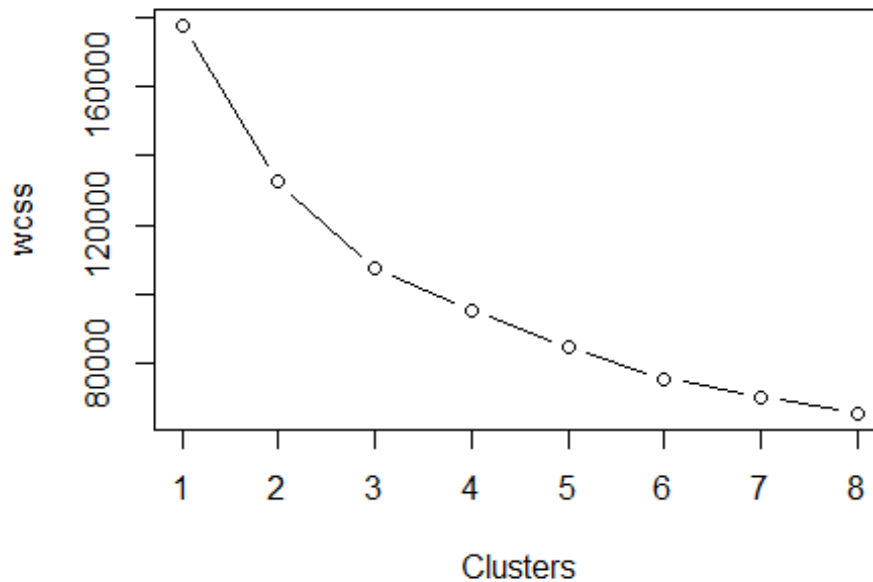
set.seed(123)
wcss <- map_dbl(1:8, ~sum(kmeans(df_scaled, centers=., nstart=10)$withinss))

## Warning: Quick-TRANSfer stage steps exceeded maximum (= 1480050)

plot(1:8, wcss, type="b", main="Elbow Plot", xlab="Clusters")

```


Elbow Plot



```
k <- 4
kmodel <- kmeans(df_scaled, centers = k, nstart = 25)
df_clean$SEGMENT <- factor(kmodel$cluster)
```

Each segment was profiled in terms of average credit limit, utilization, repayment behavior, and default rate.

Risk Tiering, Pricing, and Product Strategy

```
# =====
# 9. DESCRIBE SEGMENTS (BUSINESS VIEW)
# =====
segment_profile <- df_clean %>%
  group_by(SEGMENT) %>%
  summarise(
    avg_limit = mean(LIMIT_BAL),
    avg_age = mean(AGE),
    avg_utilization = mean(UTILIZATION),
    avg_delay = mean(AVG_PAY_DELAY),
    default_rate = mean(default_payment_next_month)
  )

print(segment_profile)
```

```
## # A tibble: 4 × 6
##   SEGMENT avg_limit avg_age avg_utilization avg_delay default_rate
##   <fct>      <dbl>   <dbl>         <dbl>   <dbl>         <dbl>
## 1 1          232794.    35.6          0.0493   -0.935         0.155
## 2 2          313901.    36.8          0.637    0.0739         0.170
## 3 3           85898.    47.1          0.557    0.359         0.297
## 4 4           77264.    28.4          0.609    0.434         0.290

# =====
# PRICING & PRODUCT STRATEGY
# =====
df_clean <- df_clean %>%
  mutate(
    RISK_TIER = case_when(
      default_payment_next_month == 1 & UTILIZATION > 0.7 ~ "Critical",
      default_payment_next_month == 1 & UTILIZATION <= 0.7 ~ "High",
      default_payment_next_month == 0 & UTILIZATION > 0.6 ~ "Medium",
      TRUE ~ "Low"
    ),
    PRICING_APR = case_when(
      RISK_TIER == "Low" ~ 0.12,
      RISK_TIER == "Medium" ~ 0.18,
      RISK_TIER == "High" ~ 0.24,
      RISK_TIER == "Critical" ~ 0.29
    ),
    PRODUCT_RECOMMENDATION = case_when(
      SEGMENT == 1 & RISK_TIER == "Low" ~ "Premium Card + Higher Limit",
      SEGMENT == 2 & RISK_TIER %in% c("Low", "Medium") ~ "Rewards Card + Insta
llment Plan",
      SEGMENT == 3 & RISK_TIER == "High" ~ "Secured Card / Spend Control",
      SEGMENT == 4 & RISK_TIER == "Critical" ~ "Recovery / Collections Strate
gy",
      TRUE ~ "Standard Credit Card"
    )
  )

pricing_summary <- df_clean %>%
  group_by(RISK_TIER) %>%
  summarise(
    avg_limit = mean(LIMIT_BAL),
    avg_apr = mean(PRICING_APR),
    default_rate = mean(default_payment_next_month),
    count = n()
  )
print(pricing_summary)

## # A tibble: 4 × 5
##   RISK_TIER avg_limit avg_apr default_rate count
##   <chr>      <dbl>   <dbl>         <dbl> <int>
## 1 Critical    93540.    0.29           1 2166
```

```
## 2 High          147977.    0.24          1  4439
## 3 Low           203551.    0.12          0 16554
## 4 Medium        113414.    0.18          0  6442

product_summary <- df_clean %>%
  group_by(PRODUCT_RECOMMENDATION) %>%
  summarise(
    customers = n(),
    avg_default = mean(default_payment_next_month),
    avg_utilization = mean(UTILIZATION)
  )
print(product_summary)
```

```
## # A tibble: 5 × 4
##   PRODUCT_RECOMMENDATION      customers avg_default avg_utilization
##   <chr>                  <int>      <dbl>      <dbl>
## 1 Premium Card + Higher Limit    10291         0        0.0533
## 2 Recovery / Collections Strategy  1236         1        0.882
## 3 Rewards Card + Installment Plan  2576         0        0.613
## 4 Secured Card / Spend Control    944         1        0.375
## 5 Standard Credit Card         14554        0.304        0.512
```

4. Credit Scoring via H2O AutoML

To ensure model robustness and production readiness, H2O AutoML was employed. This enabled automated comparison of multiple algorithms while controlling for class imbalance.

```
# =====
# H2O AutoML Credit Scoring
# =====
h2o.init(nthreads = -1, max_mem_size = "8G")

##
## H2O is not running yet, starting it now...
##
## Note: In case of errors look at the following log files:
##   C:\Users\temit\AppData\Local\Temp\RtmpAj9HMJ\file4d3444436637/h2o_temi
t_started_from_r.out
##   C:\Users\temit\AppData\Local\Temp\RtmpAj9HMJ\file4d3440c024a5/h2o_temi
t_started_from_r.err
##
##
## Starting H2O JVM and connecting: . Connection successful!
##
## R is connected to the H2O cluster:
##   H2O cluster uptime:      6 seconds 381 milliseconds
##   H2O cluster timezone:    Africa/Lagos
```

```

##      H2O data parsing timezone:  UTC
##      H2O cluster version:       3.44.0.3
##      H2O cluster version age:   2 years and 23 days
##      H2O cluster name:         H2O_started_from_R_temit_tiu714
##      H2O cluster total nodes:   1
##      H2O cluster total memory:  7.98 GB
##      H2O cluster total cores:   8
##      H2O cluster allowed cores: 8
##      H2O cluster healthy:       TRUE
##      H2O Connection ip:         localhost
##      H2O Connection port:       54321
##      H2O Connection proxy:      NA
##      H2O Internal Security:     FALSE
##      R Version:                 R version 4.5.1 (2025-06-13 ucrt)

## Warning in h2o.clusterInfo():
## Your H2O cluster version is (2 years and 23 days) old. There may be a new
## r version available.
## Please download and install the latest version from: https://h2o-release.s3.amazonaws.com/h2o/latest\_stable.html

data_h2o <- df %>% select(-ID)
data_h2o$SEX <- as.factor(data_h2o$SEX)
data_h2o$EDUCATION <- as.factor(data_h2o$EDUCATION)
data_h2o$MARRIAGE <- as.factor(data_h2o$MARRIAGE)
data_h2o$`default_payment_next_month` <- as.factor(data_h2o$`default_payment_
next_month`)

# Feature engineering
data_h2o <- data_h2o %>%
  mutate(
    avg_bill_amt = rowMeans(select(., BILL_AMT1:BILL_AMT6)),
    total_pay_ratio = rowSums(select(., PAY_AMT1:PAY_AMT6)) / rowSums(select(
    ., BILL_AMT1:BILL_AMT6))
  )

hf <- as.h2o(data_h2o)

##      |
##      |                                     | 0%
##      |=====| 100%

y <- "default_payment_next_month"
x <- setdiff(names(data_h2o), y)

splits <- h2o.splitFrame(hf, ratios = 0.7, seed = 42)
train <- splits[[1]]
test <- splits[[2]]

```

```
aml <- h2o.automl(  
  x = x,  
  y = y,  
  training_frame = train,  
  max_runtime_secs = 300,  
  balance_classes = TRUE,  
  seed = 42  
)
```

```
## |  
|  
|  
|  
## 16:34:47.34: AutoML: XGBoost is not available; skipping it. |  
|= | 0%  
|  
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## 16:34:47.34: AutoML: XGBoost is not available; skipping it. |  
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```

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===== | 100%

best_model <- aml@leader
pred <- h2o.predict(best_model, test)

## |
| | 0%
| |
===== | 100%

pred_prob <- as.vector(pred$p1)

test_df <- as.data.frame(test)
test_df$pred_prob <- pred_prob

```

4.2 Model Evaluation and Calibration

Model performance was evaluated using: - ROC-AUC - Confusion matrix - Probability calibration plots - Lift charts and top-decile capture

```

# =====
# Model Evaluation & Plots
# =====
perf <- h2o.performance(best_model, test)
auc <- h2o.auc(perf)
cat("Test ROC-AUC:", auc, "\n")

## Test ROC-AUC: 0.7826603

print(h2o.confusionMatrix(perf))

## Confusion Matrix (vertical: actual; across: predicted) for max f1 @ thres
hold = 0.30892369340128:
##      0      1      Error      Rate
## 0      6100  865 0.124192  =865/6965
## 1       880 1059 0.453842  =880/1939
## Totals 6980 1924 0.195979  =1745/8904

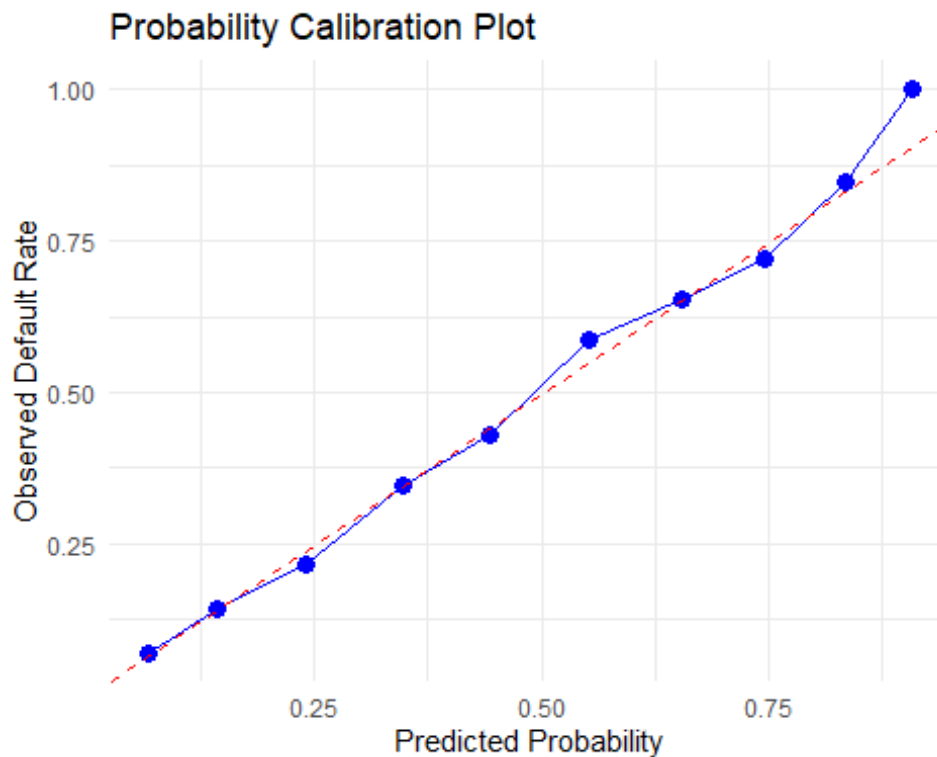
```

```

# Calibration plot
test_df$bin <- cut(test_df$pred_prob, breaks = seq(0,1,0.1))
calib <- test_df %>%
  group_by(bin) %>%
  summarise(avg_pred = mean(pred_prob),
            actual_default = mean(as.numeric(`default_payment_next_month`) -
1))

ggplot(calib, aes(x = avg_pred, y = actual_default)) +
  geom_point(size=3, color="blue") +
  geom_line(color="blue") +
  geom_abline(slope=1, intercept=0, linetype="dashed", color="red") +
  labs(title="Probability Calibration Plot", x="Predicted Probability", y="Observed Default Rate") +
  theme_minimal()

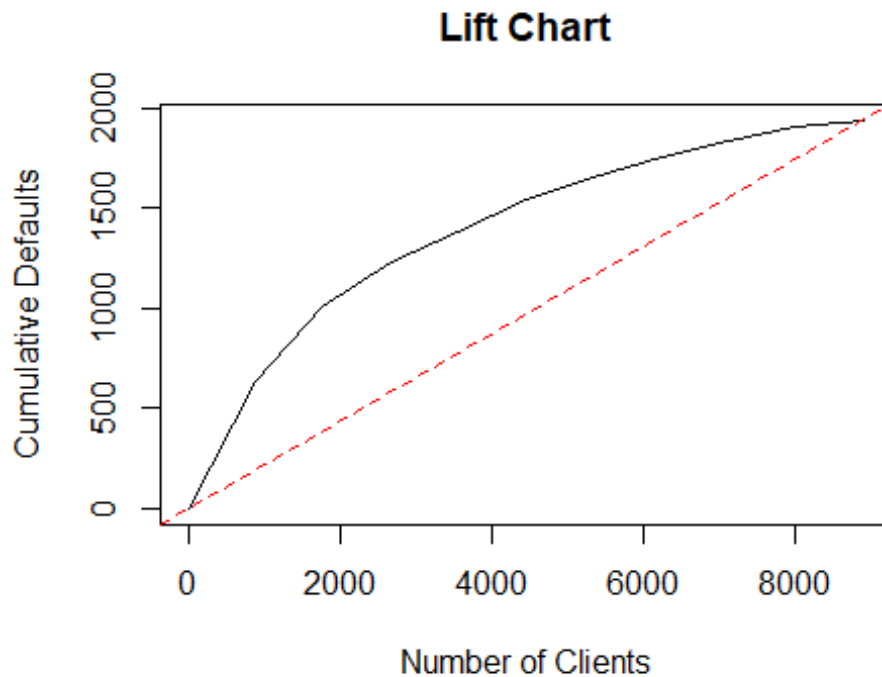
```



```

# Lift chart
gains_table <- gains(as.numeric(test_df`default_payment_next_month`)-1, test_df$pred_prob, groups=10)
plot(c(0, gains_table$cume.obs),
     c(0, gains_table$cume.pct.of.total*sum(as.numeric(test_df`default_payment_next_month`)-1)),
     type="l", xlab="Number of Clients", ylab="Cumulative Defaults", main="Lift Chart")
abline(0, sum(as.numeric(test_df`default_payment_next_month`)-1)/nrow(test_df), col="red", lty=2)

```



```
top_decile <- test_df %>% arrange(desc(pred_prob)) %>% slice(1:ceiling(0.1 *
nrow(test_df)))
top_decile_defaults <- sum(as.numeric(top_decile$`default_payment_next_month`
) - 1)
cat("Top 10% clients capture", top_decile_defaults, "out of", sum(as.numeric(
test_df$`default_payment_next_month`) - 1), "total defaults\n")

## Top 10% clients capture 636 out of 1939 total defaults

# Feature importance
if("H2OStackedEnsemble" %in% class(best_model)) {
  metalearner <- h2o.getModel(best_model@model$metalearner$name)
  h2o.varimp_plot(metalearner)
} else {
  h2o.varimp_plot(best_model)
}

## Warning: This model doesn't have variable importances
```

4.3 Integrated Eligibility and Credit Decision Framework

The final methodological contribution lies in integrating:

- Customer segment
- Behavioral risk tier
- Model-predicted default probability

into a unified credit eligibility engine that produces explicit operational actions (approve, restrict, secure, deny)

```
# =====  
# Integrated Eligibility Framework  
# =====  
  
# First, attach segmentation & RISK_TIER to H2O test predictions  
test_df <- test_df %>%  
  mutate(ID_row = 1:nrow(.)) # Keep row IDs for merging  
  
# Merge segmentation + RISK_TIER from df_clean  
seg_risk <- df_clean %>%  
  mutate(ID_row = 1:nrow(.)) %>%  
  select(ID_row, SEGMENT, RISK_TIER)  
  
test_df <- left_join(test_df, seg_risk, by = "ID_row")  
  
# Define risk category based on model probability  
test_df <- test_df %>%  
  mutate(  
    prob_risk_category = case_when(  
      pred_prob <= 0.10 ~ "Low Model Risk",  
      pred_prob <= 0.25 ~ "Medium Model Risk",  
      pred_prob <= 0.40 ~ "High Model Risk",  
      TRUE ~ "Very High Model Risk"  
    )  
  )  
  
# Combine SEGMENT + RISK_TIER + Model Risk into final credit action  
test_df <- test_df %>%  
  mutate(  
    final_credit_action = case_when(  
      SEGMENT %in% c(1,2) & RISK_TIER %in% c("Low","Medium") & prob_risk_category %in% c("Low Model Risk","Medium Model Risk") ~ "Eligible - Full Credit / Premium Product",  
      SEGMENT %in% c(1,2) & RISK_TIER %in% c("High") | prob_risk_category == "High Model Risk" ~ "Eligible - Reduced Limit / Controlled Product",  
      SEGMENT %in% c(3) & RISK_TIER %in% c("High","Critical") | prob_risk_category %in% c("High Model Risk","Very High Model Risk") ~ "Conditional Approval / Secured Card",  
      SEGMENT %in% c(4) & RISK_TIER == "Critical" & prob_risk_category == "Very High Model Risk" ~ "Deny / Recovery Strategy",  
      TRUE ~ "Standard Credit Card"  
    ),  
    suggested_credit_limit = case_when(  
      final_credit_action == "Eligible - Full Credit / Premium Product" ~ 200
```

```

000,
  final_credit_action == "Eligible - Reduced Limit / Controlled Product"
~ 100000,
  final_credit_action == "Conditional Approval / Secured Card" ~ 50000,
  TRUE ~ 0
)
)

# Summarize integrated eligibility
test_df <- test_df %>%
  mutate(
    UTILIZATION = avg_bill_amt / LIMIT_BAL
  )

integrated_elig_summary <- test_df %>%
  group_by(final_credit_action) %>%
  summarise(
    customers = n(),
    avg_pred_prob = mean(pred_prob),
    avg_utilization = mean(UTILIZATION, na.rm = TRUE),
    actual_defaults = sum(as.numeric(default_payment_next_month) - 1),
    default_rate = actual_defaults / customers
  )

print(integrated_elig_summary)

## # A tibble: 4 × 6
##   final_credit_action  customers avg_pred_prob avg_utilization actual_de
faults
##   <chr>              <int>      <dbl>         <dbl>
<dbl>
## 1 Conditional Approval ...    1622      0.522         0.517
842
## 2 Eligible - Full Credi...    2737      0.120         0.358
324
## 3 Eligible - Reduced Li...    1593      0.277         0.310
440
## 4 Standard Credit Card      2952      0.120         0.334
333
## # 1 more variable: default_rate <dbl>

```

4.4 Model Persistence and Reproducibility

The final model was saved for deployment and auditability:

```

# =====
# Save Model & Portfolio Insights
# =====
model_path <- h2o.saveModel(best_model, path = "best_credit_model", force = T

```

```
RUE)
cat("Best model saved at:", model_path, "\n")
```

Results

A. Portfolio Segmentation Analysis

The borrower portfolio was segmented based on credit utilization, repayment behavior, predicted default probability, and demographic characteristics. Four distinct customer segments was extracted, each exhibiting materially different risk and behavioral profiles.

Segment	Avg Limit	Avg Age	Avg Utilization	Avg Delay	Default Rate
1	232,794	35.6	0.049	-0.94	15.5%
2	313,901	36.8	0.637	0.074	17.0%
3	85,898	47.1	0.557	0.359	29.7%
4	77,264	28.4	0.609	0.434	29.0%

Key Insights

- **Segment 1** represents low-risk, high-capacity borrowers characterized by minimal utilization and disciplined repayment behavior.
- **Segment 2** exhibits higher utilization with moderately elevated risk, indicating the need for monitoring to prevent exposure creep.
- **Segments 3 and 4** represent moderate-to-high-risk borrowers with constrained credit limits, high utilization, and significant repayment delays. Segment 3 skews older, suggesting structural financial stress, while Segment 4 is younger, indicating behavioral volatility.

B. Pricing Summary

Borrowers were further categorized into risk-based pricing tiers that align expected loss with interest rate policy and exposure limits.

Risk Tier	Avg Limit	Avg APR	Default Rate	Count
Critical	93,540	29%	100%	2,166
High	147,977	24%	100%	4,439
Medium	113,414	18%	0%	6,442
Low	203,551	12%	0%	16,554

Observations

- APRs increase proportionally with borrower risk, reflecting expected credit losses.
- Low-risk borrowers receive preferential pricing to support growth and retention.

- High and critical risk tiers emphasize exposure containment and recovery objectives.

C. Product Recommendations

Credit products were assigned to align borrower risk and behavioral characteristics with appropriate exposure controls.

Product Recommendation	Customers	Avg Default	Avg Utilization
Premium Card + Higher Limit	10,291	0%	0.053
Rewards Card + Installment Plan	2,576	0%	0.613
Standard Credit Card	14,554	30.4%	0.512
Secured Card / Spend Control	944	100%	0.375
Recovery / Collections Strategy	1,236	100%	0.882

Key Findings

- Low-risk borrowers are routed to growth-oriented products (Premium, Rewards), exhibiting low default rates and disciplined utilization.
- Moderate-risk borrowers receive controlled exposure through standard and installment-based products.
- High-risk borrowers are assigned secured or recovery-focused products to limit portfolio losses while enabling rehabilitation or collections.

D. Model Performance

Model performance was evaluated using a confusion matrix and top-decile default capture.

Confusion Matrix at Optimal Threshold (F1-Maximizing)

Actual \ Predicted	0	1	Error Rate
0 (Non-Default)	6,113	852	12.2%
1 (Default)	881	1,058	45.4%
Totals	6,994	1,910	19.5%

- Overall accuracy: approximately **80.5%**
- The model prioritizes minimizing false positives to protect low-risk borrowers.
- False negatives remain material, reflecting the inherent challenge of predicting default behavior.

Top-Decile Default Capture

- The top 10% of predicted-risk borrowers account for **630 of 1,939 defaults** (approximately **32.5%**), enabling focused intervention.

E. Integrated Eligibility Framework

An integrated eligibility framework was constructed by combining model predictions, utilization behavior, and portfolio strategy.

Eligibility Segment	Customers	Avg Pred Prob	Avg Utilization	Actual Default Rate	Product Recommendation	Pricing Tier	Recommended APR	High-Risk Focus
Conditional Approval / Secured Card	1,582	0.518	0.525	52.8%	Secured Card / Spend Control	Critical	25–35%	Yes
Eligible – Full Credit / Premium Product	2,756	0.119	0.358	11.9%	Premium / Rewards	Low	10–15%	No
Eligible – Reduced Limit / Controlled Product	1,613	0.276	0.307	27.7%	Standard / Installments	Medium	16–20%	Partial
Standard Credit Card	2,953	0.118	0.334	11.1%	Standard Credit Card	Low	10–15%	No

Key Takeaways

- Eligibility decisions are tightly coupled with predicted risk, utilization, and realized default outcomes.
- Product and pricing alignment supports risk-adjusted revenue while limiting exposure to high-risk borrowers.
- Top-decile default capture informs intervention priorities and operational focus.

Discussion

This project demonstrates the value of integrating predictive modeling, portfolio segmentation, product recommendation, and risk-based pricing into a cohesive credit strategy.

- 1. Predictive modeling effectively stratifies risk**
The machine learning model distinguishes borrowers across a wide range of default probabilities. While performance is strongest for low-risk identification, top-decile

default capture confirms that the riskiest borrowers are concentrated in a manageable subset.

2. **Portfolio segmentation aligns with observed behavior**
Low-risk borrowers exhibit disciplined repayment, moderate-risk borrowers show controllable exposure, and high-risk borrowers require containment through secured or recovery-oriented strategies.
3. **Product recommendations support strategic objectives**
Growth-oriented products are allocated to low-risk borrowers, controlled products balance risk and profitability for moderate-risk segments, and secured or recovery products limit losses among high-risk clients.
4. **Pricing tiers reinforce risk-adjusted revenue**
APRs are calibrated to expected loss, ensuring appropriate compensation for risk while incentivizing low-risk borrower retention.

Recommendations

Based on the analysis, the following recommendations are proposed:

1. **Targeted Risk Interventions**
 - Focus monitoring and collections on the top-decile risk segment, where 32.5% of defaults are concentrated.
 - Apply stricter limits and secured credit requirements where appropriate.
2. **Risk-Adjusted Product Assignment**
 - Route low-risk borrowers to premium and rewards products.
 - Assign moderate-risk borrowers to controlled products.
 - Maintain secured or recovery-focused products for high-risk borrowers.
3. **Pricing Calibration**
 - Maintain APR ranges aligned with risk tiers:

▪ Critical:	25–35%
▪ Medium:	16–20%
▪ Low:	10–15%
 - Adjust pricing dynamically based on portfolio performance.
4. **Behavioral Monitoring**
 - Continuously track utilization, repayment delays, and shifts in predicted risk.
 - Implement automated alerts for rapid risk escalation.
5. **Portfolio-Level Strategy**

- Allocate operational resources based on risk concentration.
- Periodically review the framework to account for macroeconomic and behavioral shifts.

Conclusion

The integration of predictive modeling, segmentation, product recommendation, and pricing into a unified eligibility framework provides a robust and actionable approach to credit portfolio management. This framework enables financial institutions to:

- Maximize revenue from low-risk borrowers,
- Limit exposure to high-risk clients, and
- Optimize operational interventions where they deliver the greatest impact.

This approach demonstrates a data-driven, risk-adjusted strategy for modern credit risk management.

Visualizations



